

# Guaranteed-Friend Lunch Seating: Simulation Methodology, Solver Pipeline and Results

Isaac Lin\*

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## Abstract

This document is a complete description of the simulation behind the guaranteed-friend lunch seating demonstration: the synthetic cohort generator, the submission rules and the exact coercion-capability screen, the four-stage seating solver (pod construction, swap repair, group-swap simulated annealing, hinted CP-SAT), the random-seating baseline, the statistics reported, and the validation, determinism and timing guarantees. Every number quoted here is read from the committed trace files (`docs/data/*.json`) by `paper/make_tables.py`, so the text cannot drift from the data. In the honest scenario every submitter sits with a listed friend in all 16 rotations, 99.32% of submitters sit with exactly one listed friend on average, and students meet 73.43 distinct schoolmates over the year against 72.31 under random seating. A six-student coalition captures a table in 16/16 rotations with no defenses, is split into pairs (mean cluster 2.06, 0/16 intact) under the min-4 rule, and is flagged before rotation 1 by the coercion screen with zero honest students affected.

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\*Code and traces: <https://github.com/IsaacLin247/lunch-seating-visual>. Interactive site: <https://isaacclin247.github.io/lunch-seating-visual/>.

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## 1 Problem setting

One lunch period seats grades 11 and 12 together:  $N = 257$  students ( $n_{11} = 132$  juniors,  $n_{12} = 125$  seniors) at 40 tables, 17 of which seat 7 and 23 of which seat 6, so  $\sum_t c_t = 119 + 138 = 257 = N$  and every seat is filled. Tables are reassigned every two weeks, giving 16 rotations per school year. The school alternates two seating *states*: in a *mixed* rotation any student may sit at any table; in a *same-grade* rotation juniors sit only with juniors and seniors only with seniors. In same-grade rotations the tables are partitioned by grade so that capacities match exactly: juniors receive 12 seven-seat and 8 six-seat tables ( $84 + 48 = 132$ ), seniors 5 seven-seat and 15 six-seat tables ( $35 + 90 = 125$ ). In the implementation tables 0–16 seat 7 and tables 17–39 seat 6; the junior tables in same-grade rotations are 0–11 and 17–24, the senior tables 12–16 and 25–39 (`sim/solver.py:table_layout`). The committed traces start with a mixed rotation and alternate from there.

**Mechanism.** Each student confidentially submits a list of at least four names, or no list at all (the *min-4-or-none* rule); at least two of the names must be in the student’s own grade so that the guarantee below is satisfiable in same-grade rotations. Write  $L_i$  for student  $i$ ’s submitted list ( $|L_i| \leq 8$  in all scenarios) and call  $i$  a *submitter* if  $L_i \neq \emptyset$ . In every rotation the solver must seat each submitter with at least one listed peer; in same-grade rotations only same-grade listed peers count, since cross-grade peers cannot share a table. Among all such seatings the solver prefers ones in which each student has exactly one listed peer at the table (one familiar face, everyone else new) and in which pairs of students who have already shared a table this year are not seated together again.

**Notation.**  $x_{i,t} \in \{0, 1\}$  indicates that  $i$  sits at table  $t$ . For a seating,  $s_i$  is the number of  $i$ ’s (eligible) listed peers at  $i$ ’s table.  $m_{ab}$  counts the rotations so far in which  $a$  and  $b$  shared a table, and  $\alpha_{ab}$  counts those rotations for pairs in which at least one of the two listed the other ( $\alpha_{ab} = m_{ab}$  for a listed pair and 0 otherwise). All history is year-to-date and is updated after every rotation.

## 2 Synthetic cohort

No real student data is used anywhere. Students are identified as S001–S257; the first 132 are juniors. The generator lives in `sim/generator.py` and is deterministic in its seed.

### 2.1 Popularity, grade bias and reciprocity

Each student receives a popularity weight  $\pi_i \sim \text{LogNormal}(0, \sigma^2)$  with  $\sigma = 0.6$ , drawn from `numpy.random.default_rng(seed)`. Each student has  $K = 8$  true friends. Friendships are directed and are sampled in a random order over students: when student  $i$  is processed, the

number of open slots is  $K$  minus the friends  $i$  already has, and that many distinct targets  $j \neq i$  are drawn without replacement with probability proportional to

$$w_{ij} = \pi_j \cdot \begin{cases} 9 & \text{if } g_j = g_i \\ 1 & \text{otherwise,} \end{cases}$$

which is the 9:1 within-grade bias. Each newly sampled edge  $i \rightarrow j$  is reciprocated ( $j \rightarrow i$  is added) with probability  $p$  provided  $j$  still has an open slot. Because reciprocated edges fill slots that would otherwise have been sampled, every student ends with exactly  $K$  friends when  $\mu < 1$  (Section 2.2). The reciprocation probability is calibrated deterministically: the sampler is run for  $p \in \{0.25, 0.30, \dots, 0.75\}$  with the same sub-seed and the run whose measured reciprocity (fraction of directed edges whose reverse also exists) is closest to 0.5 is kept. For the committed traces the measured reciprocity is 0.490 and the within-grade fraction of friendships is 0.904.

## 2.2 Latent friend groups

The function

```
make_cohort(seed, mu, omega, group_size_dist=(6,12), cross_grade_group_frac=0)
```

adds community structure. Students of each grade are shuffled and cut into groups whose sizes are drawn uniformly from  $\{6, \dots, 12\}$ ; with probability `cross_grade_group_frac` a group is instead assembled from both grades' pools. A remainder smaller than 6 is folded into the last group that already holds students of that grade. With probability  $\omega$  a student additionally joins one secondary group chosen uniformly among the other groups. A student's *group pool* is the union of their group(s) minus themselves.

When  $\mu > 0$ , each of the  $K$  friendship draws is assigned to the group pool with probability  $\mu$  (a  $\text{Binomial}(K, \mu)$  count per student) and to the background otherwise. Group draws are taken without replacement from the pool with probability proportional to popularity; background draws use  $w_{ij}$  above with the student's own group pool excluded, so that the in-group share of friendships is  $\mu$  in expectation and  $\mu$  is identifiable from the data. If the pool is exhausted the unfilled group draws fall back to the background when  $\mu < 1$ ; at  $\mu = 1$  exactly they are dropped, so groups are fully isolated and members of small groups have fewer than 8 friends (never fewer than 5 with these sizes). Reciprocation is applied afterwards exactly as above. Unit tests (`tests/test_generator.py`) check that the measured in-group fraction is within  $\pm 0.05$  of  $\mu$  for  $\mu \in \{0.3, 0.6, 0.8\}$ , that the secondary-group frequency is within  $\pm 0.05$  of  $\omega$ , that the within-grade fraction at  $\mu = 0$  is unchanged, and that  $\mu = 1$  yields isolated cliques. The committed traces use  $\mu = 0$ ,  $\omega = 0$ , in which case the generator is bit-identical to the plain grade-biased network.

## 2.3 The planted friend group

Six mid-popularity juniors are planted as mutual true friends: without latent groups they are drawn from the juniors whose popularity rank lies between the 35th and 65th percentile; with latent groups they are the six members closest to median popularity of the junior group whose median popularity is nearest the junior median. Each keeps their  $K - 5 = 3$  most preferred outside friends. These six (S015, S030, S055, S110, S115, S121) are the coalition in the gaming scenarios and appear with honest lists in every other scenario.

## 2.4 Submitted lists

Honest students submit their true friends in preference order, filtered by `apply_rules`: duplicates and self are removed; a list with fewer than four names counts as no submission; if fewer than two names are in the student's grade, same-grade acquaintances are added with probability proportional to popularity; if the list is still shorter than four, further names are added from the

grade-biased background. With  $K = 8$  and the 9:1 bias no honest list in the committed traces needed padding (every student has at least four same-grade friends). A configurable fraction of non-submitters exists in the code (`nonsubmit_frac`) and is 0 in all committed scenarios.

The coalition’s lists are replaced according to the scenario:

- **k=1 chain** (no defenses): member  $c_k$  lists only  $c_{k+1 \bmod 6}$ . Every member’s single listed peer must be at their table, so any feasible seating places the whole cycle at one table.
- **4-of-5** (min-4 rule): each member lists four of the five others. The omitted names form two 3-cycles,  $c_0$  omits  $c_1$ ,  $c_1$  omits  $c_2$ ,  $c_2$  omits  $c_0$  and likewise for  $c_3, c_4, c_5$ . Restricted to a triple the lists then form a directed 3-cycle, so each triple is *closed* (every member has a listed peer inside it) and a seating that keeps a triple together gives each member exactly one familiar face at no penalty. This is the wiring that leaves the solver the most penalty-free ways of keeping members together.
- **screens on**: the coalition first submits the 4-of-5 wiring; the screen of Section 3 returns the group; the returned students resubmit their honest lists (five co-members plus three outsiders, rules applied), and the year is played with those.

### 3 Submission screens

The screens run once, before rotation 1, on the submitted lists (`sim/screens.py`).

#### 3.1 Candidate detectors

Two cheap detectors produce candidate sets. The *insular detector* computes, for every submitter  $i$ , the list-closure  $\text{cl}(i)$ : the set reachable from  $i$  by repeatedly following listed names, including  $i$ . Closures of size 2 to 12 are candidates. The *boundary detector* joins two submitters when their lists share at least three names, takes connected components of size at least two, and reports a component  $C$  whose *outward* names,  $\bigcup_{c \in C} L_c \setminus C$ , number at most three. Components larger than 12 are discarded. Both detectors are kept as functions for comparison but neither flags on its own.

#### 3.2 Exact coercion-capability test

**Definition 1.** A set  $A$  is *closed* if every member of  $A$  has at least one listed peer inside  $A$ . A candidate  $S$  admits a closed bipartition if  $S = A \uplus B$  with  $|A|, |B| \geq 2$  and both  $A$  and  $B$  closed.

The flagging rule (stated on Tab 4 of the site) is:  $S$  is flagged exactly when it admits no closed bipartition. A closed part is a set of students that can be seated by itself with every member’s guarantee met; if no such split exists the hard constraint forces all of  $S$  onto one table (capture if  $|S| \leq 7$ , infeasibility otherwise). Because  $|S| \leq 12$  the test is decided exactly by enumerating the  $2^{|S|-1}$  subsets  $A$  that contain the smallest member of  $S$  (bitmask enumeration, closedness checked on the mask, short-circuit on the first valid split); sets of size at most three are flagged automatically since two parts of size at least two cannot exist.

Students who belong to a flagged set are *flagged*; the set *returned* for diversification additionally includes every member of a candidate set that contains a flagged student, so the whole group is asked to diversify rather than only the coercive core. The screen also reports min-4-or-none violations (lists with one to three names) separately.

#### 3.3 Properties and outcome

The property tests in `tests/test_screens.py` exercise wirings defined inline: complete digraphs of size 4–12 never flag (every split is closed); pure chains (each member lists exactly the next around a cycle) always flag; hub wirings (everyone lists a common hub and the next member of a chain, the hub lists member 1) flag; circulant lists (each member lists the next  $k \geq 2$  around a circle) do not flag because an interleaved split exists; a shared-outsider wiring (six members each

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**Algorithm 1** One rotation

---

```
1:  $\sigma \leftarrow \text{PODGREEDY}$  ▷ Stage 1
2:  $\text{SWAPREPAIR}(\sigma)$  ▷ Stage 2
3:  $\sigma \leftarrow \text{ANNEAL}(\sigma, 300\,000)$  ▷ Stage 3, full-year history
4: for up to 6 attempts while a submitter is friendless do
5:    $\text{SWAPREPAIR}(\sigma); \sigma \leftarrow \text{ANNEAL}(\sigma, 150\,000)$ 
6: end for
7:  $\sigma' \leftarrow \text{CPSAT}(\text{hint} = \sigma)$  ▷ Stage 4, previous-rotation pairs
8: if  $\sigma'$  found, has no friendless submitter, and  $C(\sigma') \leq C(\sigma)$  then  $\sigma \leftarrow \sigma'$ 
9: end if
10: if a submitter is still friendless then
11:    $\sigma \leftarrow \text{CPSAT}(\text{hint} = \sigma)$  with six times the budget; raise an error if none
12: end if
13: return  $\sigma$ 
```

---

list the next three co-members and one common outsider) is detected by the boundary detector and passes; a pair listing only each other flags. On the synthetic cohort the honest lists produce no candidates at all, an honest cohort with  $\mu = 1$  produces 30 candidate kernels (one per isolated group) none of which flags, and the  $k=1$  chain flags as a single kernel of six.

**Proposition 1.** *Any closed  $S$  with  $|S| \leq 8$  in which every member lists at least four names inside  $S$  admits a closed bipartition.*

*Proof.* At least  $4|S|$  directed edges lie among  $\binom{|S|}{2} \leq 28 < 4|S|$  pairs when  $|S| \leq 8$ , so some pair  $\{x, y\}$  is mutual;  $A = \{x, y\}$  is closed, and every  $b \in S \setminus A$  lists at least four names in  $S$  of which at most two are in  $A$ , so  $B = S \setminus A$  is closed and  $|B| \geq 2$  for  $|S| \geq 4$ .  $\square$

Consequently a six-student coalition that complies with the min-4 rule can never form a coercive kernel, which is why the solver is free to scatter it (Section 7). For the 4-of-5 wiring the six-set candidate accordingly passes with the split S015, S030, S055 | S110, S115, S121. The boundary detector, however, also reports the two omission triples as separate clusters (their pairwise overlaps are three names and their outward names are exactly the other triple), and a triple cannot be bipartitioned into parts of size two; in the committed traces the triple S110, S115, S121 is flagged (Table 4), the kernel containing it returns all 6 members, and no honest student is affected. After resubmission the screen reports nothing.

## 4 The seating solver

One rotation is solved by `solve_rotation` in `sim/solver.py`, given the lists, the state, the year-to-date pair history  $m_{ab}$  and the previous rotation’s tables. Algorithm 1 summarises the pipeline; Sections 4.2–4.6 give the exact rules.

### 4.1 Effective lists and pair weights

For the rotation, the *effective* list of  $i$  is  $L_i$  in a mixed rotation and  $\{j \in L_i : g_j = g_i\}$  in a same-grade rotation;  $s_i$  counts effective peers at  $i$ ’s table. A pair is *listed* if either lists the other (full lists). The Stage 3 pair weight is

$$W_{ab} = 5m_{ab} + 2\alpha_{ab}, \quad \alpha_{ab} = \begin{cases} m_{ab} & \text{listed pair} \\ 0 & \text{otherwise,} \end{cases}$$

so a listed pair that has met  $m$  times weighs  $7m$  and an incidental pair  $5m$ . (This follows the stated energy literally; it charges anchor repeats more than incidental ones, which is the opposite

of the earlier heuristic “incidental repeats cost more”, and is a one-line change if  $m_{ab}$  was meant to count incidental co-seatings only.) In same-grade rotations a student may only be placed at tables of their grade; in the local-search stages swaps are drawn within a grade so this never has to be checked.

## 4.2 Stage 1: pod greedy

Students are processed in increasing order of effective list size (ties broken randomly), so the student with the fewest options is seeded first,  $i^* = \arg \min_{i \in U} |L_i \cap U|$  over the unmatched set  $U$ . Each  $i$  is paired with an unmatched listed peer  $j$ , choosing a peer who lists  $i$  back first, then the smallest  $W_{ij}$ , then the smallest list size; a peer  $j$  who does not list  $i$  back and has strictly fewer options than  $i$  is never taken, so a scarce student’s only slot is not consumed one-directionally. Everyone left becomes a singleton pod.

Pods are then merged until every submitter has a listed peer inside their own pod. Needy students are processed in increasing order of list size, and a needy  $m$  merges their pod with the pod of one of their listed peers provided the merged size does not exceed 6 (the smallest table), choosing the target pod by the key (extra familiar faces the merge creates  $\times 1$  plus  $\sum W$  across the two pods, then target size, then random). This is what assembles one-directional chains: with the  $k=1$  coalition the three pairs merge into one pod of six before any honest student can claim a seat next to a member.

Finally pods are placed largest first. For each pod every table with enough remaining seats (and the right grade) is scored by  $\sum W$  between pod members and current occupants plus 1 per list relation between them, ties broken toward emptier tables and then randomly. A pod that fits nowhere is split into singles placed in any free seat. The result fills every table exactly.

## 4.3 Stage 2: swap repair

The local-search state keeps, per table, the members and the cached table cost

$$\text{cost}(t) = \sum_{i \in t} \text{pen}_i(s_i) + \sum_{\{a,b\} \subseteq t} W_{ab}, \quad \text{pen}_i(s) = \begin{cases} 1000 & s = 0 \text{ and } i \text{ submitted} \\ \max(0, s - 1) & \text{otherwise,} \end{cases}$$

so that the total cost  $C(\sigma) = \sum_t \text{cost}(t)$  is the negative of the Stage 3 energy  $E$  up to a constant (Section 4.5); in code every term is multiplied by 10 to stay in integers. Any move exchanges a set  $G_a$  at table  $t_a$  with an equally sized set  $G_b$  at  $t_b$ ; its delta is computed exactly by re-costing the two affected tables.

For up to 20 rounds, every friendless submitter  $i$  (in random order) tries the single swap  $i \leftrightarrow x$  for every  $x \neq j$  at the table of every listed peer  $j$  of  $i$ , and applies the best one if its delta is below 500 (half a violation), which means  $i$  becomes satisfied without leaving anyone else friendless. If no single swap achieves that, the compound move  $\text{KICKANDREPAIR}(i, j)$  is tried for  $i$ ’s peers  $j$  in random order: the closure of  $i$  (Section 4.4) is forced onto  $j$ ’s table, an equally sized set of  $j$ ’s tablemates outside  $j$ ’s closure is evicted to  $i$ ’s old table (preferring students with the longest lists, who are easiest to re-seat), and every evictee or tablemate left friendless is re-seated by the best single swap toward one of their own listed peers with delta below 500; the whole compound is kept only if the total cost decreased and is otherwise undone in place. In the committed traces repair leaves no friendless submitter in any rotation, so the retry loop of Algorithm 1 never runs.

## 4.4 Group closure

Single swaps almost always break a guarantee (moving a student away from their only anchor orphans both), so the annealer moves *closed groups*. The closure of student  $a$  at table  $t$  is grown from  $\{a\}$  by two rules until stable: (i) any submitter  $x$  at  $t$  whose every anchor at  $t$  is already in the group joins it (otherwise  $x$  would be orphaned when the group leaves); (ii) any submitter in

the group with no listed peer inside the group pulls in one of their anchors at  $t$  (chosen at random). A closure larger than 7 (a whole table) is abandoned. Moving a closure therefore never leaves anyone at the origin table friendless, and the movers stay anchored among themselves whenever possible.

#### 4.5 Stage 3: group-swap simulated annealing

The energy maximised is exactly

$$E(\sigma) = \sum_i \left( 1000 \mathbf{1}[s_i \geq 1] - \max(0, s_i - 1) \right) - \frac{1}{10} \sum_{(a,b) \text{ co-seated}} (5 m_{ab} + 2 \alpha_{ab}),$$

where the first sum runs over submitters for the indicator term (non-submitters contribute a constant) and the second over pairs sharing a table in the proposed seating, with full-year history. Acceptance is Metropolis,  $P(\text{accept } \Delta E < 0) = e^{\Delta E/T}$ , with  $T$  geometric from 2.5 to 0.02 over 300 000 iterations; the code minimises  $10C$  with  $T$  scaled by the same factor, which leaves every acceptance probability unchanged. The best state seen is returned.

Each iteration draws a student  $a$  uniformly and a partner  $b$  as follows.

- With probability 0.25:  $b$  uniform in  $a$ 's grade pool (any student in a mixed rotation), and the move is the raw single swap  $\{a\} \leftrightarrow \{b\}$ .
- With probability 0.25: a listed peer  $j$  of  $a$  is drawn,  $b$  is a uniform tablemate of  $j$  other than  $j$ , and the move is the single swap  $\{a\} \leftrightarrow \{b\}$  (peer-directed: students are often proposed next to a friend).
- Otherwise:  $b$  uniform as in the first case, and the move is the closure swap  $G_a \leftrightarrow G_b$  with  $G_a = \text{cl}(a)$ ,  $G_b = \text{cl}(b)$ ; the smaller closure is padded with up to eight random draws of extra tablemates until the sizes match, and the move is skipped if they still differ.
- While the current cost still contains a violation, every 8th iteration is a targeted repair instead:  $a$  is a uniformly drawn friendless submitter,  $j$  one of their listed peers and  $b$  a tablemate of  $j$ ; with probability  $\frac{1}{2}$  the move is the closure swap and otherwise the single swap. Every 64th such iteration runs `KICKANDREPAIR( $a, j$ )` instead of a Metropolis move.

Moves whose two students share a table are skipped. A closure swap recomputes two table costs of at most seven members each, so an iteration costs about  $15 \mu\text{s}$  and the stage takes about 4 s.

#### 4.6 Stage 4: CP-SAT certify and polish

The annealed seating is passed as a complete hint to Google OR-Tools CP-SAT (v9.15). Variables and constraints use a per-student indicator encoding, with no pairwise variables for the friend counts:

$$\begin{aligned} x_{i,t} &\in \{0, 1\} \text{ for every table } t \text{ allowed for } i, & \sum_t x_{i,t} &= 1 \quad \forall i, & \sum_i x_{i,t} &= c_t \quad \forall t; \\ y_{i,t} &\leq x_{i,t}, & y_{i,t} &\leq \sum_{j \in L_i} x_{j,t}, & \sum_t y_{i,t} &\geq 1 \quad \text{for every submitter } i; \\ \sum_{j \in L_i} x_{j,t} &\leq 1 + (|L_i| - 1) z_i + |L_i|(1 - x_{i,t}) & \forall t, & \text{for } |L_i| \geq 2, \end{aligned}$$

so  $z_i = 1$  whenever  $i$  sits with two or more listed peers ( $z_i$  plays the role of  $\sum_t y_{i,t}^{(2)}$ ). For every pair  $(a, b)$  seated together in the *previous* rotation with  $W_{ab} > 0$ , a repeat indicator satisfies  $r_{ab} \geq x_{a,t} + x_{b,t} - 1$  for every table  $t$  (encoded as the clause  $r_{ab} \vee \neg x_{a,t} \vee \neg x_{b,t}$ ). The objective is

$$\min \quad 300 \sum_i z_i + \sum_{(a,b)} w_{ab} r_{ab}, \quad w_{ab} = 100 (5 m_{ab} + 2 \alpha_{ab}).$$

Hints cover every variable:  $x$  from the annealed seating,  $y_{i,t} = 1$  exactly at  $i$ 's table when  $s_i \geq 1$ ,  $z_i = \mathbf{1}[s_i \geq 2]$ ,  $r_{ab} = \mathbf{1}[a, b \text{ share a table}]$ . In a mixed rotation the model has about 21 000 Boolean variables and 31 000 constraints; in a same-grade rotation about half of that. The solver runs

with `interleave_search = true`, 16 interleaved workers and a `max_deterministic_time` of 3.5 units, which makes its result a deterministic function of the model and hint (Section 6).

CP-SAT sees only previous-rotation repeats, so an optimum of its model can be worse on the full-year energy. Its seating is therefore accepted only if it leaves no submitter friendless and its full-year cost  $C$  is not higher than the annealed seating’s; otherwise the annealed seating is kept. Table 2 records which stage won each rotation: CP-SAT typically wins early in the year, when there is little history, and the annealer later. If, against expectation, a friendless submitter survived both stages, CP-SAT is rerun with six times the budget and the pipeline raises an error rather than export an invalid rotation; neither path was exercised in the committed traces.

#### 4.7 Baseline, anchors and history update

The status-quo baseline for the same rotation is a uniformly random seating that respects the state: a random permutation of all students over the seat slots in a mixed rotation, and a random permutation within each grade over that grade’s tables in a same-grade rotation, drawn from `random.Random(f"{seed}-random-{r}")`. For each submitter the *anchor* recorded in the trace is the listed peer at their table with the fewest previous co-seatings (ties by student index); the baseline’s anchors are recorded the same way. After both seatings are fixed,  $m_{ab}$  is incremented for every pair sharing a table in the proposed seating and a separate history is kept for the baseline.

### 5 Statistics

For every rotation the trace stores, over submitters, the percentage with  $s_i \geq 1$ ,  $s_i = 1$  and  $s_i \geq 2$  for both seatings; the mean over all students of the number of distinct schoolmates they have shared a table with so far (*met*); the number of pairs seated together who had met before (*repeat pairs*); the cost breakdown of the final seating; the stage whose seating was kept; and the wall-clock solve time. For coalition scenarios it also stores the clusters of coalition members per table, the largest cluster, the mean cluster size experienced by a member,  $\sum_c |c|^2/6$  (6.0 when intact, 2.0 when in three pairs, 1.0 when fully scattered), and whether all six shared a table (*intact*). The summary gives year-level aggregates of the same quantities.

### 6 Validation, determinism and cost

Before a trace is written, `validate_trace` asserts that every table holds exactly its capacity in both seatings of every rotation, that every student appears exactly once, that states alternate, that no table mixes grades in a same-grade rotation, that every submitter has an eligible listed peer at their table and that the recorded anchor is such a peer, and that the reported  $\geq 1$  percentage is 100. The pytest suite (93 tests) re-checks these on the committed files and adds the scenario-level claims: the baseline seats 5–40% of submitters with a friend (21.2% on average), the proposed system seats more than 90% with exactly one, the undefended coalition is intact and holds a table in all 16 rotations, the min-4 coalition is intact in at most one rotation with mean cluster below 3 and largest cluster at most 3, the screened scenario flags only coalition members and returns all six, and the honest scenario flags nobody. It also enforces the site’s copy rules (no em dashes anywhere under `docs/`, Tab 4 equations present verbatim and in display mode, fewer than 150 words of prose).

Every random choice is seeded: the cohort from `seed` and fixed sub-seeds, the per-rotation local search from `seed×1000 + r`, the baseline as above, and CP-SAT through its deterministic interleaved search. Re-exporting a scenario reproduces the committed file byte for byte apart from the timestamp and timing fields; this was verified for the honest and min-4 scenarios. The full export of four scenarios took 568 s on an 8-core machine, with a mean of 8.9 s and a maximum of 12.4 s per rotation (Stage 3 about 4 s, Stage 4 about 3–7 s including model construction, Stages 1–2 under 0.1 s).



**Table 1:** Headline statistics per scenario (seed 7, 16 rotations). “ $\geq 1$ ” and “exactly one” are percentages of submitters; “met” is the mean number of distinct schoolmates a student has shared a table with by rotation 16; cluster statistics describe the six coalition members.

| Scenario               | $\geq 1$ (min) | exactly one<br>mean | min  | met   | met,<br>random | intact | avg.<br>cluster |
|------------------------|----------------|---------------------|------|-------|----------------|--------|-----------------|
| Honest                 | 100.0          | 99.32               | 97.7 | 73.43 | 72.31          | –      | –               |
| Coalition, no defenses | 100.0          | 99.34               | 98.0 | 71.38 | 72.31          | 16/16  | 6.00            |
| Coalition, min-4 rule  | 100.0          | 99.25               | 97.7 | 73.05 | 72.31          | 0/16   | 2.06            |
| Coalition, screens on  | 100.0          | 99.32               | 97.7 | 73.43 | 72.31          | 0/16   | 1.60            |

**Table 2:** Honest scenario, rotation by rotation. “2+” counts submitters with two or more listed peers at their table; repeat pairs are pairs seated together who had already shared a table this year (proposed / random); “phase” is the pipeline stage whose seating was kept; time is the wall-clock solve time of the rotation.

| rot. | state | $\geq 1$ | exactly one | 2+ | met  | met, rnd | repeats (prop./rnd) | phase  | time (s) |
|------|-------|----------|-------------|----|------|----------|---------------------|--------|----------|
| 1    | mixed | 100      | 100.0       | 0  | 5.5  | 5.5      | 0 / 0               | cpsat  | 4.9      |
| 2    | same  | 100      | 100.0       | 0  | 10.9 | 10.8     | 0 / 17              | cpsat  | 7.4      |
| 3    | mixed | 100      | 99.2        | 2  | 16.3 | 16.1     | 15 / 20             | anneal | 9.1      |
| 4    | same  | 100      | 99.6        | 1  | 21.5 | 21.1     | 33 / 66             | anneal | 8.0      |
| 5    | mixed | 100      | 100.0       | 0  | 26.7 | 26.1     | 36 / 47             | anneal | 9.3      |
| 6    | same  | 100      | 99.6        | 1  | 31.6 | 30.8     | 71 / 99             | anneal | 7.9      |
| 7    | mixed | 100      | 99.6        | 1  | 36.4 | 35.6     | 79 / 85             | anneal | 9.4      |
| 8    | same  | 100      | 100.0       | 0  | 41.1 | 40.0     | 105 / 135           | anneal | 8.2      |
| 9    | mixed | 100      | 99.6        | 1  | 45.6 | 44.7     | 116 / 108           | anneal | 10.3     |
| 10   | same  | 100      | 99.2        | 2  | 49.8 | 48.8     | 164 / 179           | anneal | 8.8      |
| 11   | mixed | 100      | 99.2        | 2  | 54.2 | 53.1     | 140 / 138           | anneal | 11.1     |
| 12   | same  | 100      | 98.8        | 3  | 58.1 | 56.9     | 195 / 218           | anneal | 9.3      |
| 13   | mixed | 100      | 97.7        | 6  | 62.3 | 61.0     | 169 / 177           | anneal | 10.5     |
| 14   | same  | 100      | 98.8        | 3  | 66.0 | 64.8     | 220 / 214           | anneal | 9.0      |
| 15   | mixed | 100      | 99.6        | 1  | 70.1 | 68.8     | 178 / 180           | anneal | 10.3     |
| 16   | same  | 100      | 98.0        | 5  | 73.4 | 72.3     | 275 / 257           | anneal | 8.3      |

## 7 Results

Table 1 gives the headline numbers, Table 2 the honest scenario rotation by rotation, Table 3 the coalition’s fate under each defense level and Table 4 the screen’s verdicts. Table 5 lists every parameter.

**Honest cohort.** The guarantee holds for every submitter in every rotation. On average 99.32% of submitters have exactly one listed friend at their table; the worst rotation is late in the year, when avoiding repeats leaves the solver less room. Students meet 73.43 distinct schoolmates over the year against 72.31 under random seating: the repeat penalties make the proposed seating slightly better than random at spreading students, while the baseline seats only 21.2% of submitters with a friend in a typical rotation.

**Coalition.** With no defenses the  $k=1$  chain is seated together in 16/16 rotations (cluster size 6 throughout), as the hard constraint leaves no alternative. Under the min-4 rule, the adversarially wired 4-of-5 lists are split into three pairs in every rotation (mean cluster 2.06, largest cluster 2, 0/16 intact) while every member still receives exactly one familiar face. With screens on, the coercion screen flags 3 members before rotation 1 and returns all 6; their honest resubmission passes, and the year then plays out like everyone else’s with the six scattered across four or five tables.

**Table 3:** Coalition outcome per rotation under the three defense levels: size of the largest cluster of coalition members at one table, and the mean cluster size experienced by a member ( $\sum_c |c|^2/6$ ).

| rot. | state | no defenses |      | min-4 rule |      | screens on |      |
|------|-------|-------------|------|------------|------|------------|------|
|      |       | largest     | mean | largest    | mean | largest    | mean |
| 1    | mixed | 6           | 6.00 | 2          | 2.00 | 2          | 1.67 |
| 2    | same  | 6           | 6.00 | 2          | 2.00 | 2          | 1.67 |
| 3    | mixed | 6           | 6.00 | 2          | 2.00 | 2          | 2.00 |
| 4    | same  | 6           | 6.00 | 2          | 2.00 | 2          | 1.67 |
| 5    | mixed | 6           | 6.00 | 2          | 2.00 | 2          | 1.33 |
| 6    | same  | 6           | 6.00 | 2          | 2.00 | 2          | 1.33 |
| 7    | mixed | 6           | 6.00 | 2          | 2.00 | 2          | 1.33 |
| 8    | same  | 6           | 6.00 | 2          | 2.00 | 2          | 2.00 |
| 9    | mixed | 6           | 6.00 | 2          | 2.00 | 2          | 1.67 |
| 10   | same  | 6           | 6.00 | 2          | 2.00 | 2          | 1.67 |
| 11   | mixed | 6           | 6.00 | 2          | 2.00 | 2          | 1.67 |
| 12   | same  | 6           | 6.00 | 2          | 2.00 | 2          | 1.33 |
| 13   | mixed | 6           | 6.00 | 2          | 2.00 | 2          | 1.67 |
| 14   | same  | 6           | 6.00 | 3          | 3.00 | 2          | 1.67 |
| 15   | mixed | 6           | 6.00 | 2          | 2.00 | 2          | 1.67 |
| 16   | same  | 6           | 6.00 | 2          | 2.00 | 2          | 1.33 |

**Table 4:** Coercion screen on the coalition’s initial submission (screens-on scenario): every candidate set, how it was detected, and the verdict.

| candidate set $S$                  | source   | verdict | closed split found                  |
|------------------------------------|----------|---------|-------------------------------------|
| S015, S030, S055, S110, S115, S121 | insular  | passes  | S015, S030, S055   S110, S115, S121 |
| S110, S115, S121                   | boundary | flagged | none                                |

**Sensitivity to community structure.** A fast run (reduced budgets) of the honest scenario with  $\mu = 0.6$ ,  $\omega = 0.3$  gave 98.3% exactly-one and 73.5 distinct schoolmates against 72.3, so latent groups in this range do not change the picture materially.

**Relation to the proposal’s earlier figures.** The proposal quoted about 96% exactly-one and 67–69 distinct schoolmates against 73–75 under random seating from an earlier prototype and cohort. This pipeline lands at 99.32% and 73.43 against 72.31. Two observations: the random baseline on this cohort is 72.31 rather than 73–75, and the present pipeline avoids repeats at least as well as random seating in every rotation but the last, so it meets marginally more people rather than fewer. Neither a stronger latent-group structure nor the literal Stage 3 weights moved these numbers toward the earlier figures.

## 8 Visualisation pipeline

The site (docs/) is static HTML, CSS and JavaScript with no build step and reads the trace files directly. Rooms are drawn on `<canvas>`: 40 tables in an  $8 \times 5$  grid, junior tables of the same-grade layout on the left half and senior tables on the right, seats on a ring around each table. On a rotation change every student flies from the old seat to the new one along a slightly bowed path, staggered by origin table, over 1.2 s with cubic easing, and the grade tints fade in or out; the animation runs at 60 frames per second for the 514 dots of the side-by-side view. Colours encode the outcome (exactly one listed friend, two or more, none). Avatars are generated in the browser from the anonymous ids with DiceBear, with an initials fallback when the CDN is unreachable. Tab 1 shows random versus proposed seating with live statistics and a cumulative distinct-schoolmates chart; Tab 2 follows one student (default S123, a mid-popularity junior) with

**Table 5:** Parameters used for the committed traces.

| group     | parameter                                  | value                      |
|-----------|--------------------------------------------|----------------------------|
| cohort    | students (grade 11 / 12)                   | 257 (132 / 125)            |
| cohort    | true friends per student $K$               | 8                          |
| cohort    | popularity $\sigma$ (lognormal)            | 0.6                        |
| cohort    | within-grade weight                        | 9.0                        |
| cohort    | measured reciprocity                       | 0.490                      |
| cohort    | measured within-grade fraction             | 0.904                      |
| cohort    | latent groups $\mu, \omega$                | 0.0, 0.0                   |
| tables    | capacities                                 | $17 \times 7, 23 \times 6$ |
| rotations | count / first state                        | 16 / mixed                 |
| stage 3   | satisfied submitter                        | 1000.0                     |
| stage 3   | extra listed peer                          | 1.0                        |
| stage 3   | repeat: $m_{ab}, \alpha_{ab}$ coefficients | 0.5, 0.2                   |
| stage 3   | temperature $T_0 \rightarrow T_1$          | $2.5 \rightarrow 0.02$     |
| stage 3   | iterations                                 | 300,000                    |
| stage 4   | weight per student with $\geq 2$ peers     | 300                        |
| stage 4   | repeat scale                               | 100                        |
| stage 4   | deterministic time budget / workers        | 3.5 / 16                   |
| run       | seed                                       | 7                          |

the delivered friend highlighted each rotation, a growing wall of people met and the same student’s year under random seating; Tab 3 replays the three coalition traces with red rings and the screen banner; Tab 4 typesets the equations of this document with KaTeX. Everything is regenerated by `python sim/export_traces.py` and deployed by pushing to the `main` branch, from which GitHub Pages serves the `docs/` folder.

## 9 Limitations

The cohort is synthetic and a single seed is committed; the statistics are point estimates for that seed rather than averages over many cohorts. The CP-SAT stage is accepted in only a minority of rotations because its model ignores older history; a model carrying the full history would need per-student repeat variables and was not attempted within the time budget. The screen enumerates bipartitions exactly only for candidates of size at most 12 and relies on the two detectors to propose candidates. The literal Stage 3 energy weighs anchor repeats above incidental ones, as discussed in Section 4.

## A Trace file schema

Each scenario is one JSON file (`docs/data/<scenario>.json`, about 160 KB, no pretty-printing):

```
{ "config": { scenario, seed, n, n11, n12, K, rotations, firstState,
               tableCapacities[40], sameGradeTableGrade[40], rules, coalitionMode,
               coalition[], weights, network, solver, screen?, yearSolveSeconds },
  "students": [ { "id": "S001", "grade": 11}, ... ],
  "listed": [ ["S014", ...], ... ], // submitted lists actually solved
  "listedInitial": [...], // screened scenario: the flagged submission
  "hero": "S123",
  "rotations": [ { "idx", "state", "tables", "tablesRandom",
                   "anchors", "anchorsRandom", "stats", "coalition?", "coalitionRandom"?
                 }, ... ],
  "summary": { pctGe1Min, pctExactly1Mean, pctGe1RandomMean, meanDistinctMetFinal,
               meanDistinctMetFinalRandom, coalitionIntactRotations?,
               coalitionAvgCluster? } }
```

---

## B Reproduction

```
python -m venv .venv && . .venv/bin/activate && pip install -r requirements.txt
python sim/export_traces.py # all four scenarios, ~10 min, deterministic
pytest # 93 tests
python paper/make_tables.py && cd paper && latexmk -pdf methodology.tex
```